

Exercise set 6

Problem 0:

$n = 10$ measurements of the hardness of a new material (alloy) gave the following data:

168	185	164	182	169
181	172	185	172	180

A typical approach is to assume that the hardness measurements follow a $N(\mu, \sigma^2)$ distribution.

- Estimate μ and σ by using the standard estimators $\hat{\mu} = \bar{X}$ and $\hat{\sigma} = S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$. What is $SD(\hat{\mu})$?
- Under the assumption that the hardness measurements follow a $N(\mu, \sigma^2)$ distribution, compute a 95% confidence interval for μ and for σ . Hint: You may (or may not) have learned that we can obtain a CI for σ by $\left[\sqrt{\frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}}, \sqrt{\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}} \right]$.
- Run the Bootstrap algorithm in `bootstrap_examples.R` to estimate the standard deviation of the mean and to compute confidence intervals. Do these match well with your findings in b)? Make sure you understand every step of the code.
- Extend the Bootstrap algorithm in `bootstrap_examples.R` to estimate standard deviation and bias of the standard deviation, and compute confidence intervals for standard deviation based on the Bootstrap algorithm. Do these match well with your findings in b)?

Problem 1:

Consider the 1-dimensional random walk Metropolis Hastings function `oneDRWMH()` given in the R-example file `mcmc_1.R` (considered in movies "chap11_part1").

In typical applications of MCMC, the log-target function ($\log g(\boldsymbol{\theta})$ in the lecture notes) is often expensive to evaluate. Modify the code so that only one evaluation of log-target function per MCMC iteration is required. Check that your implementation is still correct using a target distribution of your choice.

Problem 2:

Consider the situation in Example 11.3 in Rizzo.

- Write a random walk Metropolis Hastings method with target distribution being the

posterior distribution for p , i.e. $\log g(p) = 431 \log(p) + 4 \log(1 - p)$. Find a proposal standard deviation σ that results in an acceptance rate of between 20 and 40 %.

- Check your simulated samples against the Beta(432, 5)-distribution.
- Write a random walk Metropolis Hastings method with target distribution being the posterior distribution for μ , i.e. $\log g(\mu) = 431 \log(\mu/(1 + \mu)) - 6 \log(1 + \mu)$. Find a proposal standard deviation σ that results in an acceptance rate of between 20 and 40 %.
- Check that you get the same distribution both from sampling μ directly, subsequently transforming the samples of μ back using inverse of the transformation $\mu = p/(1 - p)$.

Problem 3:

Write a random walk Metropolis Hastings function for bivariate target distribution.

Define

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

and

$$\mathbf{m} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Check your implementation using the following situations

- A bivariate normal distribution $N(\mathbf{m}, \mathbf{P}^{-1})$, i.e. $\log(\mathbf{x}) = -0.5(\mathbf{x} - \mathbf{m})^T \mathbf{P}(\mathbf{x} - \mathbf{m})$.
- A bivariate t -distribution (see wikipedia) with location vector $\mathbf{m}$, Scale matrix $\mathbf{P}^{-1}$ and $\nu = 4$ degrees of freedom, i.e. $\log g(\mathbf{x}) = -3 \log(1 + 0.25(\mathbf{x} - \mathbf{m})^T \mathbf{P}(\mathbf{x} - \mathbf{m}))$.

Check in both cases that your algorithm produces samples with the correct mean and covariance matrix. For the normal distribution, check that your x_1 -output has the correct marginal distribution, i.e. $N(m_1, (\mathbf{P}^{-1})_{1,1})$.

In both cases it is a good starting point for tuning your code to choose the proposal covariance matrix to be $2\mathbf{P}^{-1}$.

Problem 4:

Consider the simple Bayesian model:

- Parameter μ with prior distribution $\mu \sim N(0, 10^2)$
- Parameter τ with prior distribution $\tau \sim \text{Exp}(1)$.
- Observations $\mathbf{y} = [y_1, \dots, y_n]$ where each $y_i | \mu, \tau \sim N(\mu, \tau^{-1})$ independently.

The parameter $\tau = 1/\text{Var}(y_i)$ is commonly referred to as the precision parameter. The posterior distribution of $\boldsymbol{\theta} = [\mu, \tau]$ may be written as

$$p(\boldsymbol{\theta} | \mathbf{y}) \propto \left(\prod_{i=1}^n p(y_i | \mu, \tau) \right) p(\mu) p(\tau).$$

The following function evaluates $\log g(\boldsymbol{\theta})$ associated with $p(\boldsymbol{\theta} | \mathbf{y})$ (assuming the observations are already available in a vector $\mathbf{y}$):

```
lprob <- function(theta){
  mu <- theta[1]
  tau <- theta[2]
  if(tau<0.0) return(-1e100) # log target effectively - infinity for negative tau
  s.dev <- sqrt(1.0/tau)
  loglike <- sum(dnorm(y,mean=mu,sd=s.dev,log=TRUE))
  mu.log.pri <- dnorm(mu,mean=0,sd=10,log=TRUE)
  tau.log.pri <- -tau # = log(exp(-tau))
  return(loglike+mu.log.pri+tau.log.pri)
}
```

For data ($n = 7$):

```
y <- c(1.2, 6.3, 5.4, 3.4, 7.5, 4.8, 1.9)
```

Using your random walk Metropolis Hastings function for bivariate target (Problem 3):

- obtain at least 10000 samples from the posterior distribution of $\boldsymbol{\theta}$.
- find the posterior mean and standard deviation of μ , τ and $s = \tau^{-1/2}$. (for s you first transform all samples of τ , and then calculate mean and standard deviation of the samples)
- make a scatterplot of μ and s together.
- Find an interval so that s is with 95% certainty within the interval.